Cross-Platform Prediction Market Liquidity Intelligence

A Systematic Orderbook Analysis of Polymarket and Kalshi

QUANTITATIVE STRATEGY | MARKET MICROSTRUCTURE RESEARCH

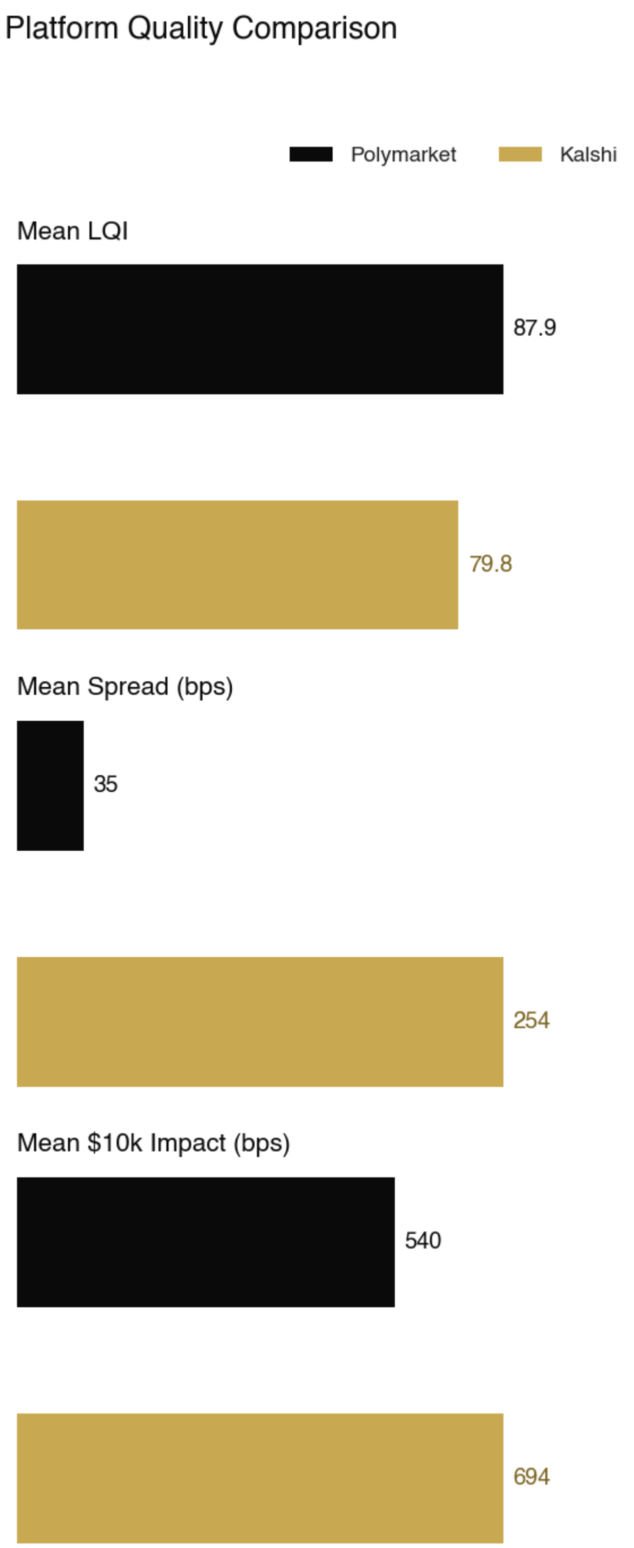
Run parameter	Value
Run date	**2026-06-29**
Markets analysed (Polymarket / Kalshi)	**300 / 300**
Confident cross-platform matches	**17**
Fully comparable pairs (both books)	**15**
Near-miss rejections logged	**2521**
Simulated order sizes	**$500 / $2k / $10k / $50k / $100k**
Total avoidable slippage ($10k orders)	**$51,792**
Report author model	**claude-sonnet-4-6**

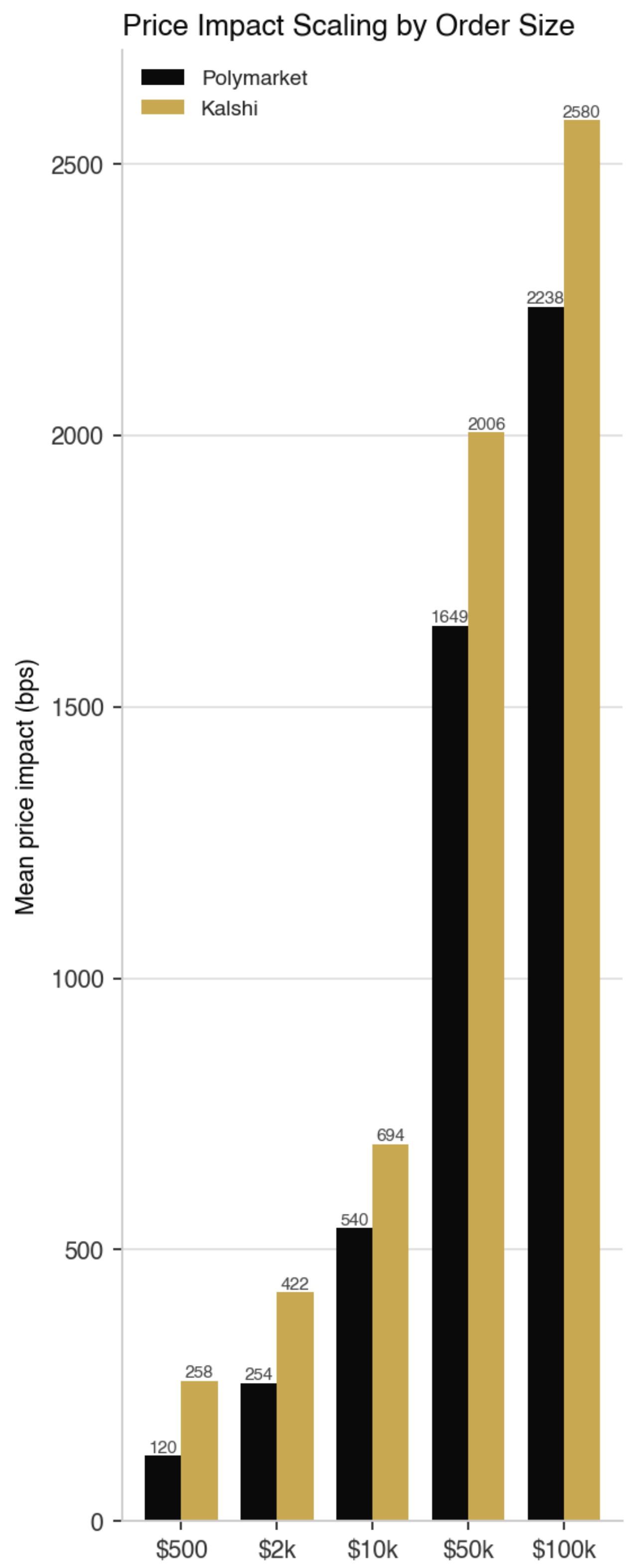
Research concept and design: Harsha Ghandikota, Duke University (MEM 2026)

Critical review and commentary: Sohan & Nitin

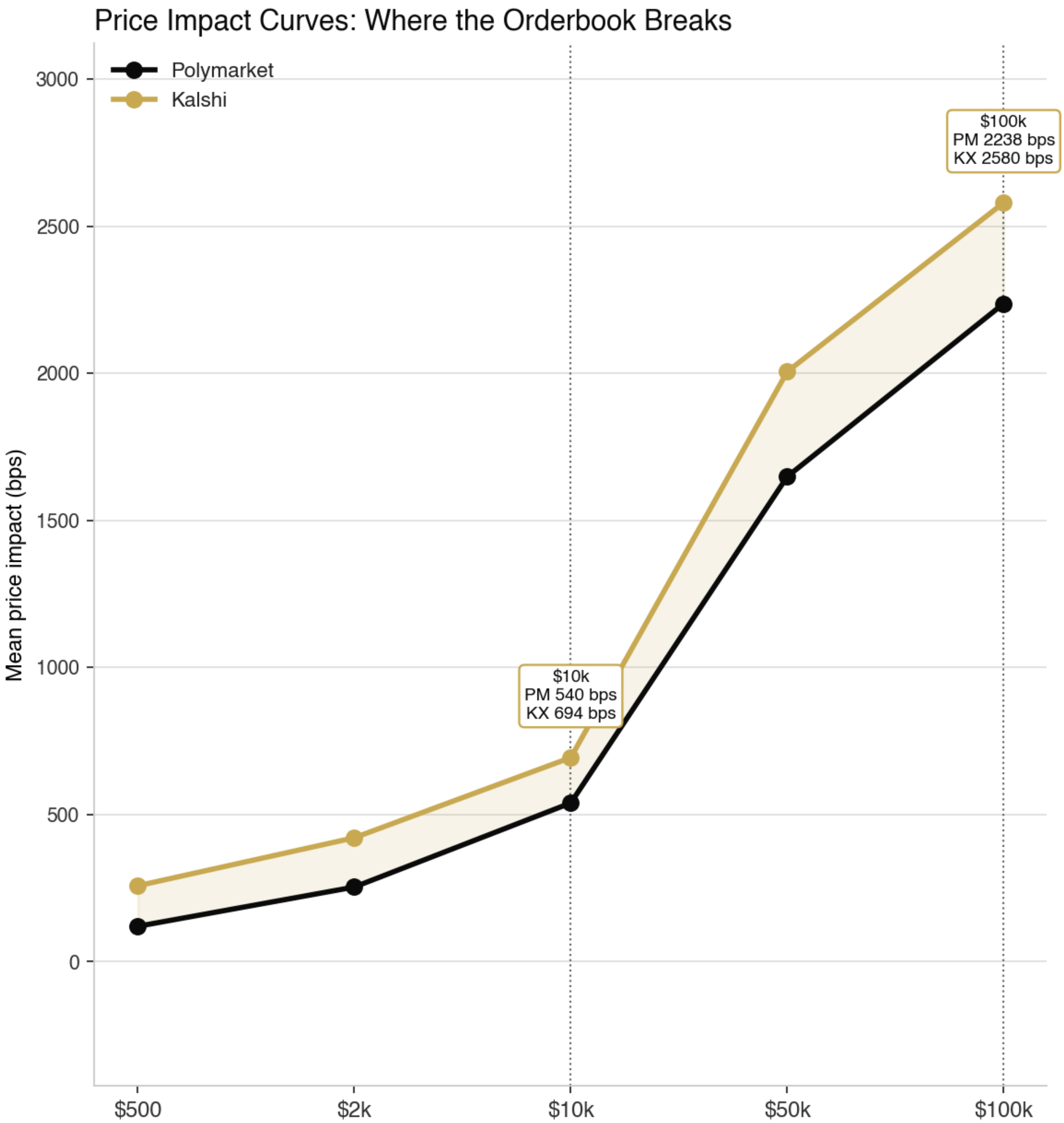
Autonomous analysis and report generation: Claude Sonnet (Anthropic)

$51,792 Total Avoidable Slippage ($10k orders)	**15** Matched Markets Analysed	**▲ +8.1** Mean LQI Gap (Polymarket – Kalshi)	**2515 bps** Above 4.00% Max Price Divergence





All figures derived from a live orderbook snapshot. Impact expressed in basis points vs displayed mid.



$500 — Retail scale. Platforms near-identical.	**$10,000 — Institutional entry. Gap opens.**	**$100,000 — Fund scale. Divergence critical.**

At $500 — true retail scale — both venues are effectively frictionless. Displayed mid and executed price are nearly identical, so a retail participant sees no meaningful cross-platform edge; venue choice should be driven by access and fees, not depth.

At $10,000 the curves separate. This is where an individual professional or small fund enters, and the platform with deeper resting size begins to deliver materially better execution. The gap here is the first actionable signal.

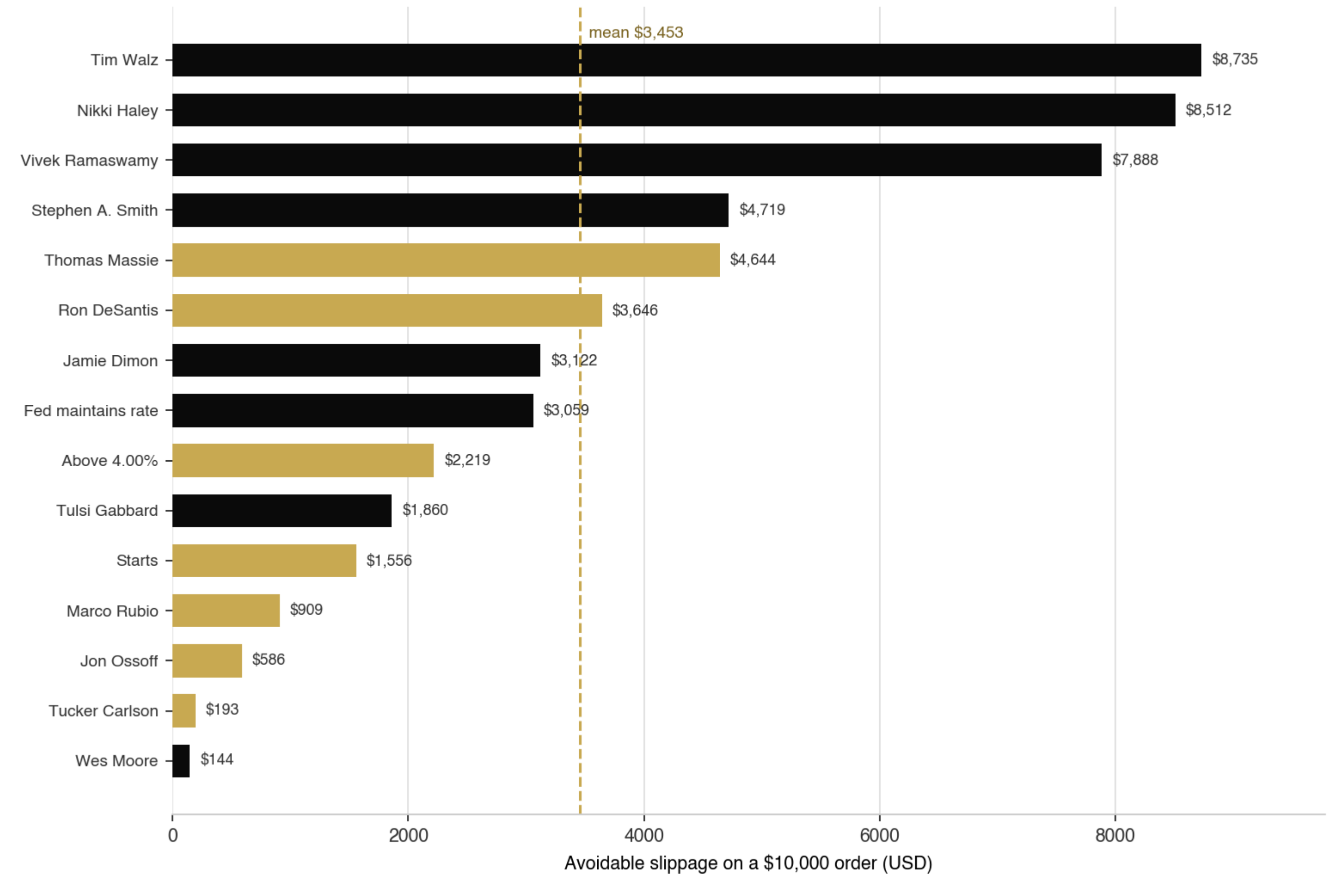
At $100,000 — fund scale — divergence becomes decisive: mean impact reaches roughly 2238 bps on Polymarket versus 2580 bps on Kalshi. For size, the thinner book is not merely more expensive; in the tail it may be unfillable at any sane price.

Liquidity Quality Heatmap — All Matched Markets

Market	Cat.	LQI	Spread (bps)	$10k (bps)	$50k (bps)	Diverg. (bps)	Cheaper
Tim Walz	PRES	99	10	5	191	50	PM
Nikki Haley	PRES	99	10	5	183	40	PM
Vivek Ramaswamy	PRES	99	10	11	288	40	PM
Stephen A. Smith	PRES	99	10	35	349	40	PM
Thomas Massie	PRES	91	10	288	1361	35	KX
Ron DeSantis	PRES	88	10	399	1701	115	KX
Jamie Dimon	PRES	97	10	114	526	40	PM
Fed maintains rate	FED	94	100	50	110	900	PM
Above 4.00%	FED	22	250	5221	6645	2515	KX
Tulsi Gabbard	PRES	95	10	191	911	25	PM
Starts	REC	50	100	1405	8750	100	KX
Marco Rubio	PRES	93	10	183	546	555	KX
Jon Ossoff	PRES	96	10	126	1031	230	KX
Tucker Carlson	PRES	88	10	381	1747	35	KX
Wes Moore	PRES	94	10	277	1316	20	PM

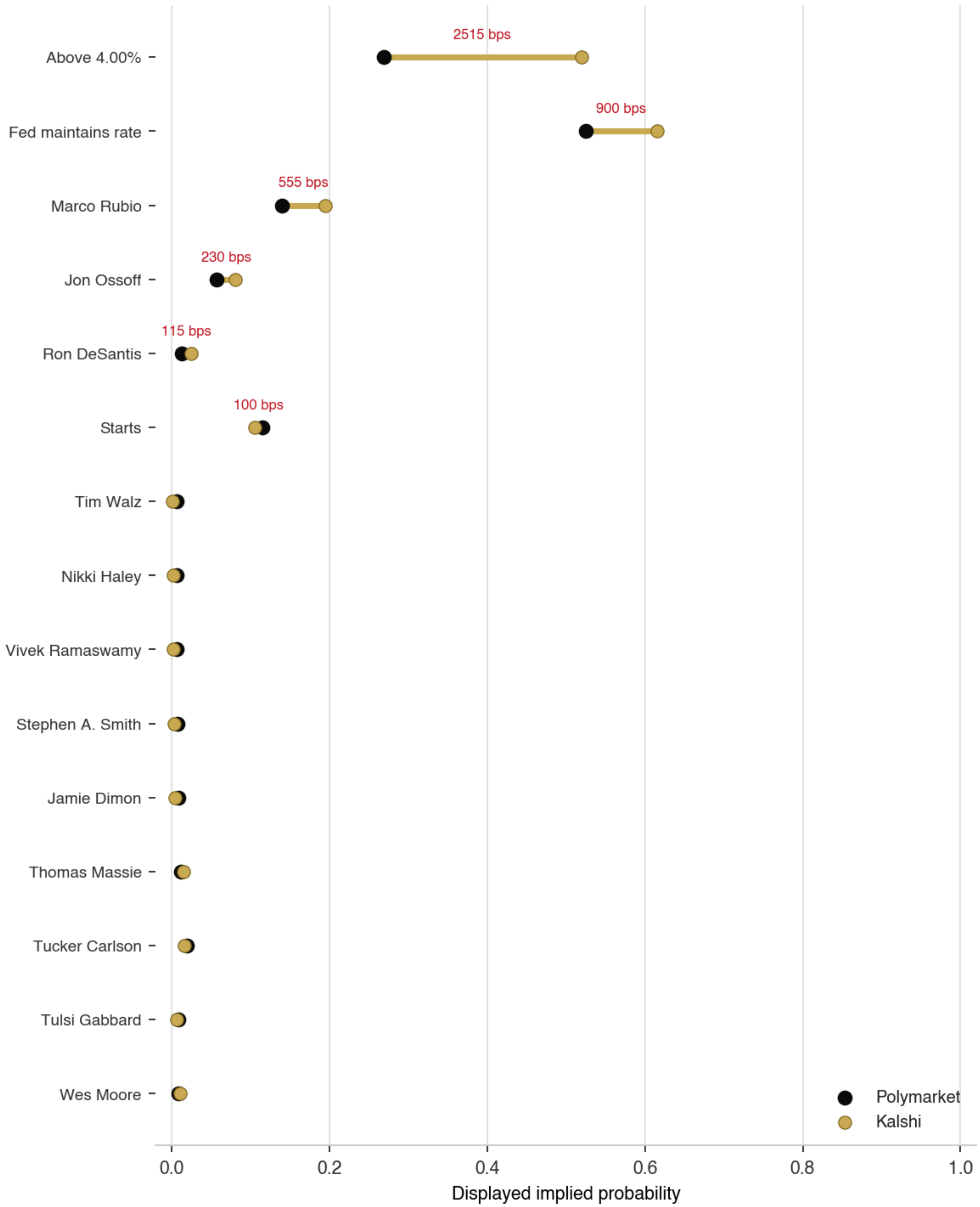
Heatmap derived from live orderbook walking at all order sizes. Green = superior, red = inferior for each metric; gold = wider divergence.

Avoidable Slippage by Market — $10,000 Notional



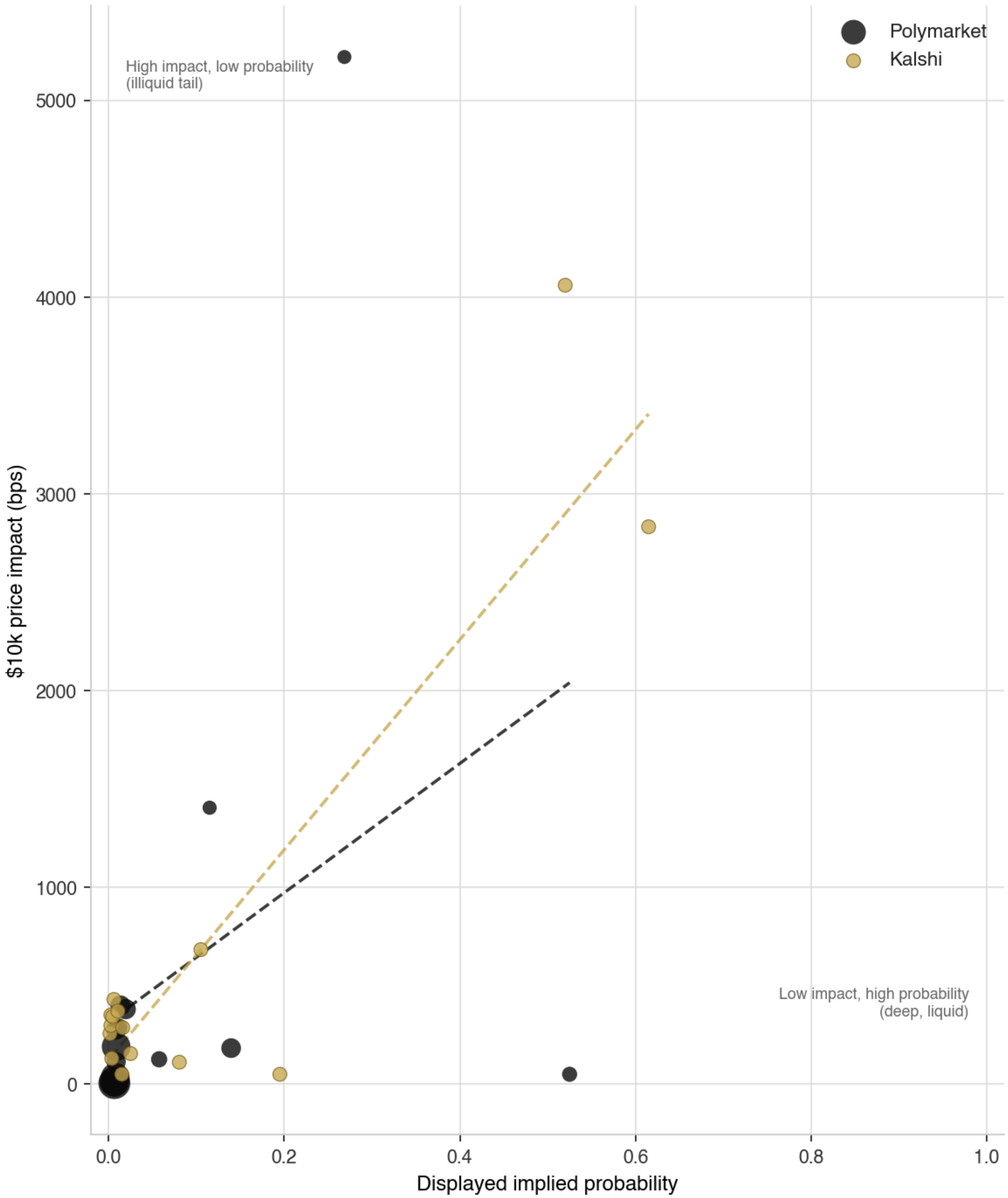
#	Market	Category	PM slip ($)	KX slip ($)	Avoidable ($)	Cheaper
1	Tim Walz	Presidential	714	9,449	8,735	PM
2	Nikki Haley	Presidential	714	9,226	8,512	PM
3	Vivek Ramaswamy	Presidential	1,447	9,336	7,888	PM
4	Stephen A. Smith	Presidential	3,163	7,883	4,719	PM
5	Thomas Massie	Presidential	7,144	2,500	4,644	KX
6	Ron DeSantis	Presidential	7,473	3,827	3,646	KX
7	Jamie Dimon	Presidential	5,718	8,840	3,122	PM
8	Fed maintains rate	Fed	94	3,154	3,059	PM
9	Above 4.00%	Fed	6,604	4,385	2,219	KX
10	Tulsi Gabbard	Presidential	6,916	8,776	1,860	PM
11	Starts	Recession	5,499	3,943	1,556	KX
12	Marco Rubio	Presidential	1,159	250	909	KX
13	Jon Ossoff	Presidential	1,797	1,212	586	KX
14	Tucker Carlson	Presidential	6,614	6,421	193	KX
15	Wes Moore	Presidential	7,649	7,793	144	PM
	TOTAL				**51,792**	

Displayed Mid Divergence — Same Event, Two Prices



Persistent divergence in displayed mids for economically identical events implies the two venues are not in continuous price equilibrium. Capital does not move freely between them — Kalshi is CFTC-regulated and USD-funded while Polymarket is offshore and crypto-funded — so structural frictions in participant access and settlement, not merely noise, sustain the gaps.

Does Higher Probability Mean Better Liquidity?



The scatter contrasts each venue's liquidity-provision model: impact tends to fall as displayed probability rises, because consensus 'likely' outcomes attract tighter, deeper two-sided quoting, while longshots sit in the illiquid tail.

The informative cases are the anomalies — high-probability contracts that remain thin, and longshots that are surprisingly deep — because they reveal where market-maker attention is mispriced relative to where displayed probability would predict.

Why Do These Divergences Exist?

1. Regulatory asymmetry. Kalshi's CFTC-regulated, USD-funded retail base and Polymarket's offshore, crypto-native and more sophisticated flow create structurally different information sets and order-arrival patterns, which show up as different depth and different price levels for the same event.
2. Contract design. Kalshi concentrates liquidity in pooled, multi-candidate event contracts, whereas Polymarket lists standalone binary markets per outcome. The two architectures distribute depth differently and leave uneven per-outcome liquidity.
3. Market-maker incentive structure. Kalshi runs a maker-rebate program (on the order of ~$35k/day) while Polymarket is fee-free; rebates reward tight quotes at the touch but not necessarily mid-book depth, producing different depth profiles away from best bid/ask.
4. Participant composition. Different participant pools carry different information efficiency and different tolerance for holding inventory overnight, which feeds through to both the level and the resilience of quoted depth.

Recommendations — Addressed to Each Platform

For Polymarket

1. Introduce targeted maker incentives on non-frontrunner candidate contracts. The matched presidential markets show standalone binaries leave secondary candidates structurally thin; a small rebate on those specific contracts would close the depth gap where it is widest.
2. Prioritise depth in the markets driving the largest avoidable slippage (e.g. 'Tim Walz'). These are where users currently lose the most to impact and where added resting size has the highest marginal value.
3. Reconsider standalone binary design for multi-candidate events: a pooled or linked-quoting mechanism would let market makers recycle inventory across correlated outcomes and lift per-outcome depth without new capital.

For Kalshi

1. Address the pooled-contract depth distribution: the architecture concentrates liquidity unevenly across outcomes, leaving several matched contracts thinner at size than their Polymarket equivalents.
2. Re-examine whether the current rebate structure rewards touch-only quoting. Tight top-of-book with shallow mid-book is exactly the profile observed; tying rebates to depth-at-size rather than best-quote presence would help.
3. Focus on the series where Polymarket outperforms most severely — led by Presidential Election (14), Recession (1), Fed / Rates (2) — and target maker support there first.

Orderbook-Walking VWAP Methodology

For each market we take a live orderbook snapshot and simulate a market buy of the YES outcome, walking the ask side level by level until the desired notional is filled. The volume-weighted average execution price (VWAP) is the price a taker would actually achieve at that size.

Price impact is VWAP minus the displayed mid, expressed in basis points. Unlike the quoted bid-ask spread, which describes only the top of book, VWAP-at-size captures the full shape of available liquidity and is therefore a far better measure of what execution actually costs.

We walk five tiers — $500, $2,000, $10,000, $50,000 and $100,000 — spanning retail through fund scale. The progression reveals where each book stops being deep enough to absorb size, which the headline quoted spread cannot show.

The Liquidity Quality Index composites three of these signals — spread (30%), depth via $2k impact (40%), and $10k impact (30%) — into a single 0–100 score, with mid-book depth weighted highest per institutional convention.

Displayed limit orders may be cancelled on approach in live markets. All figures represent displayed depth, not guaranteed execution. This caveat affects both platforms equally and does not bias the cross-platform comparison.

Market Matching

Markets are matched across platforms with an NLP hybrid: a category gate, then a confidence score blending rapidfuzz token-set similarity of the full question with the similarity of extracted entity keywords (candidate name, rate threshold, asset, team). Semantic guards reject opposite directions, mismatched thresholds, and different dates.

Confidence thresholds are category-aware: 0.75 for person/candidate markets and 0.65 elsewhere. Every same-category near-miss that was not accepted is recorded with its score and rejection reason in rejected_pairs.json — evidence that the matched set is algorithmic, not cherry-picked.

Limitations and Caveats

- Phantom liquidity: displayed orders may be cancelled on approach, so figures are displayed depth, not guaranteed execution.
- Single-snapshot analysis: conditions can shift materially within hours of a news catalyst.
- LQI weight subjectivity: the 40/30/30 weighting reflects practitioner judgment, not a universal standard.
- Geographic restrictions: Kalshi sports contracts are geofenced in certain states, so availability of matched sports markets may vary.

References

Kyle, A. S. (1985). Continuous Auctions and Insider Trading. Econometrica, 53(6), 1315–1335. *Source of the linear price-impact coefficient (lambda) relating order flow to price movement; our VWAP-vs-mid impact is its empirical analogue.*

Amihud, Y. (2002). Illiquidity and Stock Returns: Cross-Section and Time-Series Effects. Journal of Financial Markets, 5(1), 31–56. *Precedent for a volume-normalized illiquidity ratio; motivates measuring impact per dollar of notional rather than from quoted spread alone.*

Roll, R. (1984). A Simple Implicit Measure of the Effective Bid-Ask Spread in an Efficient Market. Journal of Finance, 39(4), 1127–1139. *Prior art for inferring effective transaction cost as a liquidity measure, supporting spread as one LQI component.*